Virtual Vahana: Phase 1

System Overview: A safety-first, manual-primary Advanced Driver Assistance System (ADAS) implemented in the CARLA Simulator, featuring sensor-fused Autonomous Emergency Braking (AEB), Adaptive Cruise Control (ACC), and Lane Keeping Assist (LKA).

1. ADAS System Architecture

The system architecture is designed around a continuous, synchronous perception-to-actuation loop running at 20 Hz (0.05s fixed delta).

* Perception Layer: A forward-facing Radar (100m range, 25° horizontal FOV, 10° vertical FOV) fused with ego-vehicle telemetry (speed, location). The radar strictly filters ground noise (altitude < 0.05) to prevent false positives.

* Logic & Processing Layer: Calculates kinematic data using 1D Kalman Filters for noise reduction. It computes the Time-to-Collision (TTC) and evaluates Lane Waypoint deviations.

* Arbitration Layer (The "Brain"): A hierarchical state machine that compares driver inputs (WASD) against safety thresholds. It enforces mutual exclusivity (preventing simultaneous throttle and braking) and applies exponential smoothing to prevent mechanical jerk.

* Actuation & HMI Layer: Outputs validated carla.VehicleControl commands and renders real-time telemetry and alerts to the Pygame dashboard.

2. Implemented ADAS Features

* Lane Keeping Assist (LKA): Provides a gentle, proportional steering nudge to keep the vehicle centered in the lane. It automatically yields to the driver if a hard manual steering input (> 10\%) is detected.

* Adaptive Cruise Control (ACC): When the driver applies throttle, ACC monitors the forward gap. If the vehicle breaches the dynamic 2.2-second following distance, ACC autonomously restricts throttle and applies braking to match the lead vehicle's speed.

* Multi-Stage Autonomous Emergency Braking (AEB): Detects imminent collisions with vehicles or pedestrians. It features a two-stage intervention: a Forward Collision Warning (FCW) and a full hardware-override Emergency Brake.

* Speed Governor: Hard-caps the vehicle speed at 40 km/h as per urban simulation safety parameters.

* Smart Vehicle Lighting: Automatically triggers left/right blinkers based on steering angle and illuminates brake lights during manual or autonomous deceleration.

3. Feature Logic Explanation

Time to Collision (TTC) Calculations

Standard TTC relies purely on relative velocity, which can fail or return ∞ if a pedestrian or vehicle is completely stationary. To solve this, our system calculates a stabilized Closing Speed that factors in the ego vehicle's speed (v_{ego}):

This ensures that if we are approaching a stationary object at high speed, the system recognizes the closing rate. TTC is then calculated as:

(Note: TTC is only calculated if $v_{closing} > 0.5$ m/s to prevent division by zero or micro-noise triggers).

System Threshold Values

* Speed Limit: 40.0 km/h.

* Dynamic ACC Gap: $d_{safe} = \max(v_{ego_mps} \cdot 2.2, 12.0)$. This enforces a strict 2.2-second rule, with a hard minimum gap of 12.0 meters.

* FCW Threshold (SAFE_TTC): 3.5 seconds or absolute distance < 12.0 meters.

* AEB Threshold (CRITICAL_TTC): 1.8 seconds or absolute distance < 6.0 meters (Panic threshold for sudden pedestrian crosswalk events).

* Control Smoothing (α): 0.4 (Applies a low-pass filter to actuator commands to mimic real-world hydraulic braking latency).

Arbitration Logic (Safety Priority Hierarchy)

The Arbitration Layer evaluates commands from bottom to top, where the highest safety risk holds the ultimate override authority.

* Level 0 (Manual Input): The driver's raw WASD inputs are registered.

* Level 1 (Governor): If speed $\ge$ 40 km/h, manual throttle is forced to 0.0.

* Level 2 (ACC Intervention): If distance < dynamic_safe_dist, throttle is scaled down proportionally to the gap ratio. If the gap ratio drops below 50%, partial braking is applied.

* Level 3 (AEB Override - Highest Priority): If TTC < 1.8s or distance < 6.0m, the system declares an emergency. It forces Throttle = 0.0 and Brake = 1.0, completely stripping longitudinal control from the driver.

* Mutual Exclusivity: Before sending commands to CARLA, the arbitrator ensures if brake > 0.1, throttle = 0.0 to prevent engine stalling/fighting.

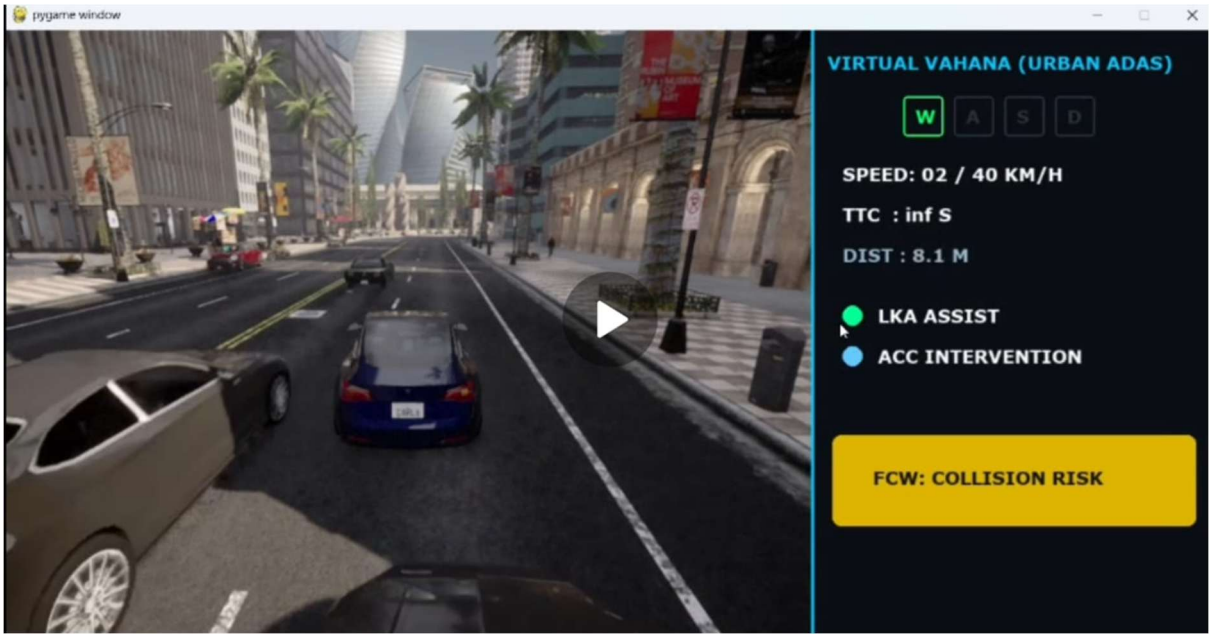
4. Dashboard Screenshots / Video & UI Visualization

(Instructions for your submission: Use OBS Studio or Windows Game Bar (Win+G) to record your screen. Take 3 distinct screenshots of the Pygame window to include in your report).

5. Visual UI Layout Description:

The Custom HMI (Human-Machine Interface) is split into two panes. The left pane (800x600) streams the live RGB camera feed from the vehicle. The right pane (400x600) is a dedicated Cyber-Dark telemetry dashboard.

Alert Visualization Guidelines

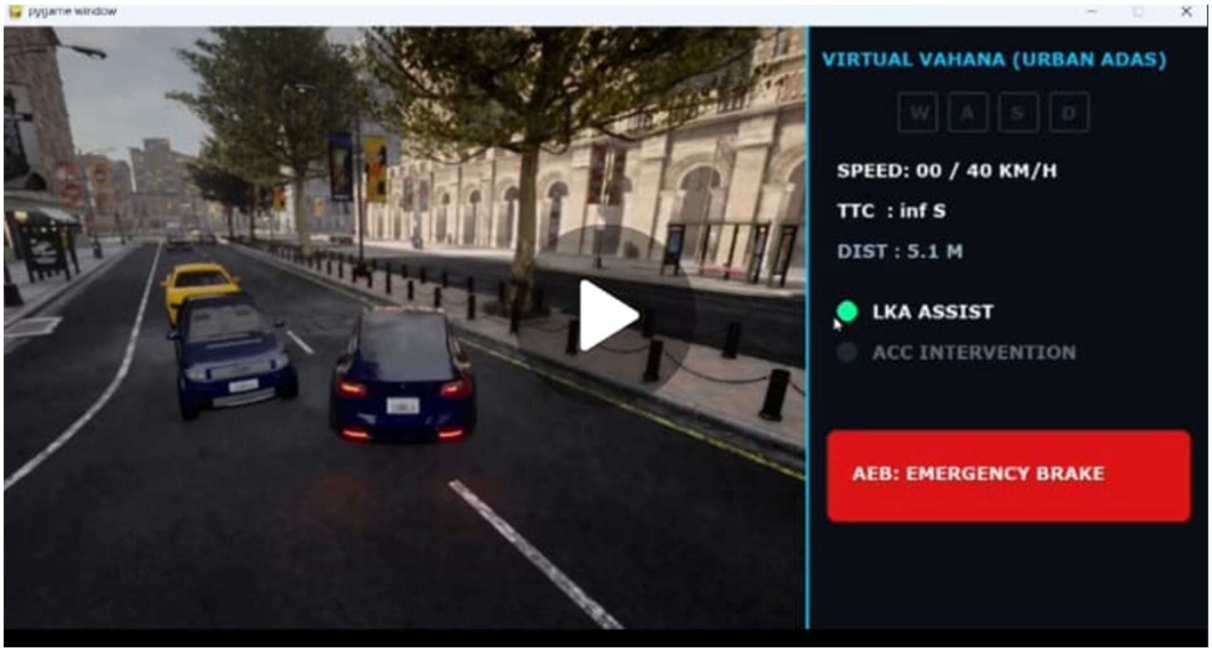


Normal Driving & LKA

* What it shows: The driver pressing 'W' (highlighted in Green on the Input Matrix). Speed is under 40 km/h. The "LKA ASSIST" lamp is glowing green.

*ACC Intervention

* What it shows: The vehicle approaching the lead Audi TT. The "ACC INTERVENTION" lamp turns blue. The SAFE GAP variable dynamically updates on the screen.



*AEB & FCW Alerts (Pedestrian Crossing)

* What it shows: A pedestrian in the crosswalk. The TTC drops below 1.8s. The UI flashes a large, bright Red Warning Box reading "AEB: EMERGENCY BRAKE". The Input Matrix shows the driver is still holding 'W', but the telemetry proves the car has stopped—proving the arbitration logic successfully overrode the human driver.